

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer 2025

Course:BTech

Branch :Mechanical Engineering

Semester :IV

Subject Code & Name: BTMEC402_Y19 Theory of Machines-I

Max Marks: 60

Date:03/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.

2. Question No. 1 will be compulsory and include objective-type questions.

3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.

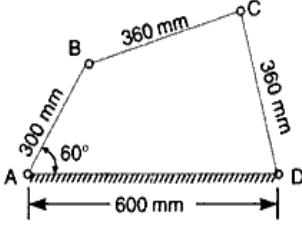
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

5. Use of non-programmable scientific calculators is allowed.

6. Assume suitable data wherever necessary and mention it clearly.

				(Level/CO)	Marks	
Q. 1	Objective type questions. (Compulsory Question)				12	
1	In a reciprocating steam engine, which of the following forms a kinematic link ?				CO1	1
	a. cylinder and piston	b.piston rod and connecting rod	c. crank shaft and flywheel	d.flywheel and engine frame		
2	The motion of a piston in the cylinder of a steam engine is an example of				CO2	1
	a. completely constrained motion	b.incompletely constrained motion	c. successfully constrained motion	d. none of these		
3	The component of the acceleration, parallel to the velocity of the particle, at the given instant is called				CO3	1
	a. The component of the acceleration, parallel to the velocity of the particle, at the given instant is called	b.tangential component	c. coriolis component	d. none of these		
4	The direction of linear velocity of any point on a link with respect to another point on the same link is				CO3	1

	a.parallel to the link joining the points	b.perpendicular to the link joining the points	c. at 45° to the link joining the points	d.none of these		
5	The angle of inclination of the plane, at which the body begins to move down the plane, is called				CO3	1
	a. angle of friction	b. angle of repose	c. angle of projection	d. none of these		
6	The frictional torque transmitted by a disc or plate clutch is same as that of				CO4	1
	a. flat pivot bearing	b.flat collar bearing	c. conical pivot bearing	d.trapezoidal pivot bearing		
7	The frictional torque transmitted by a cone clutch is same as that of				CO4	1
	a. flat pivot bearing	b.flat collar bearing	c. conical pivot bearing	d.trapezoidal pivot bearing		
8	The brakes commonly used in railway trains is				CO4	1
	a. shoe brake	b. band brake	c. band and block brake	d.internal expanding brake		
9	The brake commonly used in motor cars is				CO4	1
	a. shoe brake	b.band brake	c. band and block brak	d.internal expanding brake		
10	The size of a cam depends upon				CO5	1
	a. base circle	b.pitch circle	c. prime circle	d.pitch curve		
11	The angle between the direction of the follower motion and a normal to the pitch curve is called				CO5	1
	a. pitch angle	b.prime angle	c. base angle	d.pressure angle		
12	he balancing of rotating and reciprocating parts of an engine is necessary when it runs at				CO6	1

	a. slow speed	b. medium speed	c. high speed	d. none of these		
Q. 2	Solve the following.					12
A)	Explain inversion of four chain mechanism				CO1	6
B)	In a pin jointed four bar mechanism, as shown in Fig.1.1, $AB = 300$ mm, $BC = CD = 360$ mm, and $AD = 600$ mm. The angle $BAD = 60^\circ$. The crank AB rotates uniformly at 100 r.p.m. Locate all the instantaneous centre and find the angular velocity of the link BC  Fig 1.1				CO2	6
Q.3	Solve the following.					12
A)	Explain Laws of Static Friction and Laws of Solid Friction				CO3	6
B)	Derive torque transmission capacity Flat Pivot Bearing on uniform pressure basis.				CO3	6
Q. 4	Solve Any Two of the following.					12
A)	Explain with neat sketch multiple clutch				CO4	6
B)	Explain Simple Band Brake				CO4	6
C)	Explain with neat sketch Epicyclic-train Dynamometer				CO4	6
Q.5	Solve Any Two of the following.					12
A)	Write short note on classification of follower				CO5	6
B)	Explain terms base circle, pitch point, lift, pressure angle, trace point, pitch curve				CO5	6
C)	A cam is to be designed for a knife edge follower with the following data : Cam lift = 40 mm during 90° of cam rotation with simple harmonic motion, Dwell for the next 30° , During the next 60° of cam				CO5	6

	rotation, the follower returns to its original position with simple harmonic motion, Dwell during the remaining 180°, Draw the profile of the cam when the line of stroke of the follower passes through the axis of the cam shaft,		
Q. 6	Solve Any Two of the following.		12
A)	Explain . Balancing of Several Masses Rotating in the Same Plane	CO6	6
B)	Four masses m_1 , m_2 , m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are 45°, 75° and 135°. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m	CO6	6
C)	Explain Balancing of a Single Rotating Mass By a Single Mass Rotating in the Same Plane	CO6	6
	*** End ***		